Pharmacy Requirements

Feature 1 - User Account Management

F1.1:- User login

Description: System shall display a login form accessible from the home page, containing 2 fields labeled “E-mail” and “Password” and a button labeled “Login”..

Validation: To successfully log in, a user has to provide:

- An e-mail already registered with a non-disabled account, of valid e-mail form (e.g. “x@domain.com”)
- an e-mail may only have one account linked to it
- The password associated with their account
- Otherwise, error message containing invalid field and what’s wrong

F1.2:- Account registration

Description: System shall display a registration form, accessible from the login form, from the label “Don’t have an account? Register one now!”, which will contain 5 fields with labels for e-mail, phone number, username, password and confirming password and a button labeled “Register Account”.

The system shall store user passwords in encrypted form in the database.

Validation: To successfully create the account a user has to provide

- An e-mail not already linked with an account
- A password which follows the requirements:
  - -minimum 8 characters long
  - -at least an uppercase letter
  - -at least a lowercase letter
  - -letters can only be from the English alphabet
  - -at least one digit
  - -at least one special character from: !,@,#,%,^,*
  - -no other character is accepted
- The password, repeated in next field for confirming password
- A phone number, containing only digits, optional input
- Username, text containing only letters from the English alphabet and/or the character “_”, optional input
- If validation fails, the system shall display an error message indicating the invalid field and what’s wrong

F1.3:- Profile management

Description: System shall allow logged-in users to view their account information in a separate form, accessible from a button on the top bar. The form contains:

- A button with a clipboard icon, called “Order History”, which would take the user to their order history page
- The e-mail linked to the account
- Phone number linked, empty if none was provided
- Username field
- Password field, with a label underneath: “Change Password”
- Button labeled “Save changes”
- Button labeled “Cancel”

The user is allowed to modify the “Username” field (including leaving it empty, in which case the part of the e-mail before “@” will be used) and the phone number section, which has the same restrictions as when creating the account. Clicking cancel would exit the form without modifying any user information, while clicking “Save changes” would try to modify the changed fields. If validation fails an error message is displayed in a pop-up window saying which field is invalid. If only one field is invalid, but multiple were changed, none of the changes are accepted. If validation is successful, a message will be displayed saying “Changes saved successfully”.

When clicking the “Change Password”, a new form will be opened asking the user to input their old password, a new one and to retype the new password. If the 2 new passwords don’t match or the old password is incorrect or the new password does not comply with the password rules, an error message will be displayed and no changes will be made. There will also be a button labeled “Change Password”. If everything is successful, the form will close and a message saying “Changed password successfully” will be displayed.

All changes must be stored permanently in the database.

Administrators shall have access to a “Dashboard”, not visible to clients, accessible from the home page, which when clicked, will show a small menu where all the admin related feature buttons will be.

F1.4 Admin Account modification

Description: Admins shall be able to view and modify certain user information of every created account, through a separate window, accessible from the admin dashboard.

The form contains:

- Search bar which allows admins to filter the users shown through userId, username or e-mail, using certain prefixes in the search (“id:”, “username:”, “mail:”)
- The user accounts will be shown as info cards with the information: e-mail, username (blank if empty), phone number and a checkbox with the label “Admin”
- Admin accounts will have the info cards visually highlighted and the checked box greyed and unmodifiable
- An option to disable the account, only shown for client account

Validation:

- For client accounts, admins can check the admin box, which will make a warning appear, saying that this action cannot be undone
- If “Proceed” is selected, the selected user becomes an Admin and gets access to all the admin features
- If the option to disable the account is selected, a warning will appear

- If the admin decides to proceed, the user account will be marked: Disabled and will no longer be accessible by the user
- The user cannot create another account with the same e-mail

Feature 2 - Order management

Description: Allows users to place, view, modify, cancel and resubmit pharmacy orders. The system maintains order status and expiration information and validates all operations against inventory availability and medicine expiration dates.

Administrators can view all orders, complete pending orders, and undo completed orders through an order management interface.

F2.1:- Order placement

Description: When a user proceeds to checkout from the basket, the system shall display a checkout page where the user can review their selected items and choose a pickup date.

The page shall contain:

- List of medicines from the order, sorted by when they were added to the basket
- The quantities of each medicine
- Total price
- A calendar menu where the user can select a pickup date
- A button labeled “Place Order”
- A button labeled “Cancel”

The system shall automatically suggest a pickup date equal to the next day. Users may select any later date using the date selector.

Validation: To successfully place an order:

- The pickup date must be at least one day after the current day
- All the medicines have an expiration date later than the pickup date
- If the selected pickup is invalid, the system must display a message

F2.2:- Order History

Description: The system shall allow users to view their order history in a separate page, accessible from their account information tab.

The order history page shall contain the orders linked to the user’s account, displayed as info cards, each containing:

- OrderID to give to the pharmacist
- Order creation date
- Pickup date
- Order status (Complete/Incomplete)
- Order expiration date
- A button labeled “Cancel Order” for incomplete orders

Order statuses can be:

- Incomplete, no medicines/items have been taken out of the inventory and the order has to be completed by an admin

- Complete, after being approved by an admin

F2.3:- Modification of Incomplete Orders

Description: Users shall be able to modify Incomplete orders which have not expired, from the Order History page.

When clicked on an order, a list of all medicines with respective quantities will be displayed in a pop-up window, which will contain:

- An option to decrease/ increase the quantity
- An option to remove an item entirely
- A button labeled “Add item”
- A button labeled “Change Pickup Date”
- A button labeled “Save changes”
- A button to “Cancel”, which would leave the order as it was

Validation: If the “Save changes” button is pressed and there are no issues regarding the new contents, the order information will be modified permanently and a message with success will appear.

F2.4:- Order cancellation and Expiration

Description: The user shall be able to cancel any Incomplete order from the Order History tab, by clicking the “Cancel Order” button on the desired order. There will also be an option in the order menu to show expired orders using a checkbox.

When pressed:

- A pop-up window, asking for confirmation will appear with an option to cancel and one to proceed
- After confirming, the order shall be marked as “Expired” and will no longer be editable
- If the option to show them is ticked, expired orders will be shown in the list

Orders are automatically marked as “Expired” after their expiration date passes.

Expired orders shall have the option to “Cancel” replaced with the option to “Resubmit”, which will open a new checkout window with the same items where the user will have to pick a new pickup date.

Validation: Same validation rules apply as for F2.1. If validation is successful:

- New order with new orderId shall be created
- Original expired order shall remain unchanged

F2.5:- Admins and orders

Description: Admin users shall have access to an Order Management Page, accessible from the dashboard, different from their own Order History, where they will see a list of all orders with the option to filter by inputted userId and/or orderId, a checkbox for showing only incomplete orders and one for showing only orders that have not expired.

The page shall display the following information for each order:

- Order ID
- User ID
- Order status, which shall be colored differently depending on completeness
- Pickup date
- Expiration status, expired orders shall be visually highlighted

- Expiration date

Selecting an order shall display the order details, similar to users seeing their own orders in the history.

Admins shall have the following options in the order management page, for each order:

1. Undo completed order
    a. Visible only for complete orders
    b. Selecting this will try to add the medicines back into the inventory
    c. If none were expired, success
    d. If there are any that were expired, the system will show an error message, showing a list of the expired items with respective quantities with options to cancel the undo and to proceed
    e. Proceeding will add only the items that have an expiration date later than the current date, the others being thrown away for safety
    f. Success will prompt a message, the order will be marked as incomplete and expired, if it had any expired items
2. Complete order
    a. Visible only for incomplete orders
    b. Selecting this will attempt to complete the order, by checking to see if the medicines that are in the order's list are still available in the desired quantity, if the same medicines are in stock, but with different box-quantities or expiration date, the system will consider them as equivalent and try to complete the order replacing the unavailable medicines, trying to match the quantities desired
    c. Failure to meet this requirement will result in an error message, telling the admin that the inventory is insufficient, with the option to close the pop-up
    d. To fix order failures, the admin will have to either modify the order, removing the unavailable items, or adding in stock items that were desired.
    e. On success, the order shall be marked as completed, the inventory shall be updated accordingly and a message announcing success shall be shown

F2.6:- Basket

Description:

- The top bar of the page shall have a button with a cart icon that when pressed brings the user to the basket's page
- The basket is emptied if the user exits the application
- The basket detail page has a list view with all the products that the user wants to buy, displaying name, producer, price and quantity bought; price should be calculated from the product's initial price while quantity should be an input field for integers that can be modified either by typing another integer or using two buttons next to it to increment/decrement it
- next to the list view there should be a text displaying the price of all the products in the basket

- The basket detail page a button with the name “Checkout” that brings the user to the part specified in F2.1
- in the basket’s list view every product would have a button next to it with a trash can icon, that deletes the appropriate product from the basket; the user could also delete a product from the basket by modifying its quantity to 0

F2.7:- Add items from prescription (cross-team: MysteryInc)

Description: In the basket page, users have a field to input a prescription ID, together with a button called “Add Products”. This will add the products from that description into the basket of the user, or increment the ones already in the basket.

For products in the prescription that are not in stock, add one that is equivalent in terms of concentration of the active substances, with the same number of pills in the package, that is cheapest.

Validation: If the inputted prescription id doesn’t exist, write with red the text “invalid prescription ID” below the textbox.

If there aren’t any with the same number of pills, then add more of the products that are equivalent but have less pills, such that its still cheapest and the total number of pills for that product is greater than or equal to the amount on the prescription (for example, add two packs of 20 pills instead of one pack of 30 pills).

If there aren’t any drugs with the same concentration of active substances, get more packs of ones that have, approximately (with an error), the same concentration when added, and the total number of pills will be double the number of pills written in the prescription. If even then, there aren’t any items that can match it, just skip it and the pharmacist will tell you that physically.

Feature 3 - Pharmacy Management & QoL

Description: Allows users and admins to manage the pharmacy in different ways. Admins can update the inventory, view statistics and predictions, the application is sending notifications to users on different events.

F3.1:- Updating Pharmacy Inventory

Description: The application will display a page with multiple tabs, each containing a table. Below each table there will be 1 button to remove the selected entity from the table and database; and 2 buttons that open 1 sub-window each for adding/updating an entity (which updates the table and database). There will be a few labels with textboxes for the latter 2; these will concern the fields of the entities in the respective table, to add or update. On add it adds a new entity with the given values. On update it updates an existing entity given by its *ID/Name*. Admins can also have an option to toggle to view only **expired items** in the Items table. Also, admins can *search* for items by their name in the table. This will be done from a search box that is above the table.

1. The first table is the ***Items*** table.
    a. On add, you have 13 fields: name, price, category, number of pills in the package, producer, image, label, description, discount;

10th and 11th fields are for adding a substance and associated concentration, paired with a list view above it that shows the added substances thus far with their concentration, every list entry having a button next to it for deletion
12th and 13th are for adding batches, each with an expiration date and associated quantity, paired with a list view above it, that is similar to the substances one

b. On update, you have 14 fields: ID, name, price, category, number of pills in the package, producer, image, label, description, discount;
11th and 12th field are for substance and concentration, with substances displayed in a list the same way as in add
13th and 14th field are for expiration dates and quantities, with batches displayed in a list the same way as in add

c. On remove, it also removes the item from all orders

2. The second table is the ***Substances*** table:
   a. On add, you have 3 fields: the name, the intake limit and the description
   b. On update, you have 3 fields: the same as for add

Validation:

- On update, it updates only if the ID (for Items) / name (for Substances) already exists in the table, otherwise error message pops up “id doesn’t exist”
- If in add or update any of the fields are empty and mandatory, it highlights them red, and display an error “mandatory fields are empty”.
- If the fields do not meet the required *format* (for example, the quantity being an integer for Items), it will display an error message “invalid quantity”
- For adding substances to items: substance must be in database and its concentration in item must be lower than its lethal dosage, otherwise display error message with information about what’s wrong
- For adding batches to items: expiration date must be later than current date, otherwise display error message with “expiration date must be later than current date”
- Some fields are not mandatory and they have a default value
  - Mandatory fields will be: for Items, everything except the category (default: Medicine), quantity (default: 1) and substances(default: empty); for Substances: everything

F3.2:- Notifications

Description: Every user has a dropdown menu in the upper right corner of the home page that opens up, where notifications show up. A notification is one item in the dropdown menu that has title and content. The notifications are:

- **Product in stock**: received by users who have subscribed to an item’s stock alert by pressing a button on that product’s page entitled “Notify when ‘item’ is in stock”. The title of the notification is “Stock Alert”. The text body of this notification is the title of the product: name, number of pills, concentration(s) and producer. Clicking it brings the user to the product page.
- **Product expired:** this notification is received only by admins, and it’s received by admins on the day when a product has expired. The title is “Product Expired”. The

text body is of the form “Product: ID expired. Please remove it”. Clicking it brings the admin to the dashboard with products.

Validation:

- **Product in stock**: if the product’s quantity > 0 and was < 0 before, then we give the notification to all users who have it turned on for that product. You need to be logged in to subscribe
- **Product expired:** if the product’s expiration date is today or it passed, for a product. The batch of items will be deleted from the database, and subtracted from the quantity for that given item.

F3.3:- Statistics for admins

Description: Admins are able to see in a list view in their dashboard Statistics, which will be a page in their dashboard, with multiple tabs.

- The first tab will have **popularity statistics**, which will display in order the top 30 products sold in the last month, this helps the pharmacists know what supply to order next. Besides this, we will show another list side by side with this one that uses the substances and concentration in the top 20 products sold to predict the most popular products of the next month
- The second tab uses the bought products in the last month to find out the most **popular substances** (based on how many popular products contain it), shown in a list with name and details, and then predicts the most popular substances for the next month (the list views will be shown side by side, and in each there will be 30 active substances)

Validation: To successfully predict, we need to have the top 30 products sold, respectively top 30 active substances sold, i.e. if there aren’t 30 products, or, respectively, active substances in the top, because there weren’t enough products bought that month

Feature 4 - Product Catalogue

Description: The product catalogue is a functionality that deals with the showing of the products to the users and all the interactions that the user has with the items. This contains the Product Catalogue page but also the Home page.

F4.1: - Product Listing

Description: The product catalogue page shall display the products using a card-based layout. Each product card shall contain:

- An image
- The name of the product
- Product price in lei
- If an item is discounted it will show the new price and by how much it is discounted in percentages
- Stock availability indicator(described in validation)
- A button labeled “Add to Basket”
- If the Stock is 0 the “Add to Basket” button is disabled

Products shall be paginated, allowing users to navigate through multiple pages of items. Products shall be ordered by their internal product IDs by default.

Validation:

1. Products with stock smaller than 10 shall be labeled with the text “Only **x** in stock”, “**x**” being the numbers of items that are in stock.
2. Products with stock equal to 0 shall be labeled with the text “Out of stock”.
3. The systems checks if an item had a discount percentage > 0 and if this is the case it is treated as a discounted item

F4.2:- Product Search

Description: The product catalogue page shall display a search bar allowing users to search products by name. The search shall return products whose name matches or partially matches the entered query.

Validation:

1. Search shall be case-insensitive.
2. If no results are found, the system shall display the message in gray: “No products found.”

F4.3: - Product Filtering

Description: On the left side of the page there will be a sidebar that includes some checkboxes with all the available filters. The filter name is the text that appears above the selectable options of every checkbox.

Available filters are:

- Product category: the options are all the categories available
- Price range: the options are: 0 - 49 lei, 50 - 100 lei, 100 - 199 lei, 200 - 499 lei and 500+ lei
- Stock availability: the options are: in stock, low stock (< 10 products in stock)
- Discounted products: the options are discounted products and non-discounted products this is the only category where you can only select one option
- Active substances: the options are all the active substances from all the items available

Validation:

1. If no option is selected from a filter then the system shall display the results as if all the possible options for that filter were selected.
2. Stock availability filter
    - When the option selected is “in stock” the system displays all the items that have the stock > 0
    - When the option selected is “low stock” the system displays all the items that have the stock > 0 and < 10
3. Discounted products filter
    - When the option selected is “discounted products” the system displays the items that have the discounted field open
    - When the option selected is “non-discounted products” the system displays the items that have only a current price

4. If no products match the selected filters the system shall display the message: “No products match the selected filters”.

F4.4:- Product Sorting

Description: In the same sidebar there will be a dropdown menu that will have multiple sorting criterias. The user is able to select one of them at a time. (the default sorting is by item id) These criterias are price, rating, and newest products. Below this dropbox there shall be another one that has the criterias Ascending/Descending (the default option being Ascending).

Validation:

1. Only one criteria can be active at a time
2. If no sorting option is selected, products shall be displayed in the default order defined above.

F4.5: - Product Details Page

Description: When a user selects a product card, the system shall open a product details page containing:

- A large product image in the left part of the page
- Product name besides the image
- Price in lei below the product name. If the product has a discount, the original price shall be displayed in strikethrough format and beside it the discounted price and by what percentage. For example: “~~100~~ 25 lei 75% off”
- Stock quantity if it is < 10 otherwise display the text “In stock” or “Out of stock” depending of the stock number (the system shall display it below rating)
- Quantity selector
- Add to basket button besides quantity selector
- Product description the image and everything else
- If the items stock = 0 display the “Out of Stock” label

Validation: To interact with the Product Detail Page, the system shall validate the following conditions:

- The quantity selector value must be > 0
- The selected quantity must not exceed the available stock quantity. The quantity cannot be bigger than 50.
- If the product stock is 0, The “Add to Basket” button shall be disabled
- The discounted amount in percentages must be > 1
- If invalid quantity input is provided (not natural numbers) the system shall display a message “Invalid quantity selected”

F4.6: - Home page

Description: The system shall provide a Home page that acts like the main entry point of the application. The home page has a sidebar with all the categories available. If you click a category the user will be taken to the catalog with that category filter option selected. The home page will also display a topbar that will be available on every page that the user can see. The buttons are the following: **Products, Cycle tracker, Admin Dashboard** (This button is only available to users that are admins), **Login, Register**, **My account**, **Cart**,

Home. In the center of the Home page there will be a banner and an overlay with some text. The text should feature:

- The name of the pharmacy in big text
- The motto in a smaller text right below the name
- A quick description about the site with the same text size as the moto below the motto

Validation: The system shall validate the following conditions:

1. The Admin Dashboard button shall be visible only to users that are admins
2. For users not logged the system shall display the My Account button and hide the Login and Register Buttons
3. If a non-authenticated user attempts to press any button besides Products, Home, Login, Register and the cards of the products he/she will be redirected to the Login page
4. If the banner content is unavailable, the system shall display a default banner message.

F5:- Period tracker

F5.1:- Calculating cycle from data

Description: The application will also have an upside navigation bar link that is called “Cycle Tracker”. When a user (who is logged in) clicks it, a new page opens up.

If the user has never done this before, the page will only show a few labels with their associated fields: First day of your last period (which is a date that the user selects), how much did the last period last, and how many days a cycle lasts for the user on average. There is also an optional field titled “How many days before cycle do the PMS symptoms appear, if any”, which will have below it some checkboxes, “not applicable”, “1-3 days before period”, “4-7 days before period”, “7-14 days before period”. Default is “not applicable”

If the user has pressed “calculate” before, the data is stored in the database and re-calculated when the user opens the page (data is not null)

Below these fields arranged vertically, there will be a button labeled “Calculate Cycle”. Even after pressing it, this part will stay on the bottom of the page always, to recalculate if needed, above the recommended products.

After pressing it, the page updates to show a calendar with 3 months. To the right there is a button titled “Show the next 3 months”, that calculates the cycle for the next 3 months. The days when they have their period are background colored in pink, and the ones for the ovulation window are colored in light green. The ones with low fertility are colored in yellow, and the PMS days are colored in red. Below the calendar, there will be a legend: a pink square with the label “Period days” to its right, then a light green filled square with the label “Most probable ovulation days” to its right, and a yellow square with “Low fertility days” to its right, and then a red square with “PMS days” to its right.

Validation: If the user is not logged in, they cannot access this page, and will be sent to the log in page. The data have to be in valid format, i.e. the number of days has to be an integer, and the date will be inputted from a date pickup.

F5.2:- Notes for symptoms

Description: Below the information about the cycle, we also have a big textbox where the user can note anything, for example symptoms relating to their cycle to help them keep track of their health better. To the right of this, they have a “+” that, if pressed, will create another note. A note can also be a “TO-DO” note, which means that it will have a checkbox, which ,if clicked, will strike through the whole text, to mark it “done”. This checkbox will be to the left of the note, before the title of the note. A note can be deleted by clicking on the “X” to the upper right corner of a note.

Validation: The notes are written in UTF-8 format, and the user can create at most 4 adjacent notes. After that, the “+” disappears and it will only appear if the user deletes a note.

F5.3:- Recommended related items

Description: Below the notes, we have a checkbox to check if the user wants to get pregnant or not. If yes, then the recommended products at the end of the page will NOT have the contraceptives category too.

At the end of the page, we have some related items (below the calculating data): contraceptives, vitamins, pregnancy tests, in random order. The items are arranged in a horizontal list, showcasing the item thumbnail, its title, and its price, together with a discounted price if the user meets some conditions.

Validation: The products shown to the client below the calculator are in random order, and they will have the option to add them to the basket from there, below the item, which will result in their basket being updated. If the client is in their most probable ovulation window, or in their period (the date of today), give them a 20% discount on those suggested items, the same for period related items if they are in their predicted period days.